

## **An Agent-based Model of Whale Ship Strikes**

### **Introduction**

Globally, maritime shipping accounts for 90% of all trade (IMO, n.d.). According to the United Nations Conference on Trade Development, large commercial ships have increased from 11,108 to 94,000 between 1890 and 2018 (Schoeman et al., 2020). This increase in maritime traffic poses a growing threat to cetaceans and other marine species who experience minor to fatal injuries, a reduction in individual fitness, and a decrease in the overall population growth rate caused by ship strikes (Schoeman et al., 2020). Vessel collisions, or ship strikes, are any impact between any part of a watercraft and a live marine animal (Schoeman et al., 2020). Since 1948, the International Maritime Organization (IMO) has been the primary agency for regulating the safety, security, and environmental performance standards of international shipping (IMO, n.d.). The standards created by the IMO are then implemented and enforced on an international basis, however maritime routes falling into territorial waters can also be regulated by the concerned nation (IMO, n.d.; (Schoeman et al., 2020).

Regardless of their origin, maritime regulations need to be strongly informed by scientific data to ensure the implementation of adequate regulations that minimize the harmful impacts of anthropogenic activities on marine ecosystems. An international database recording cetacean ship strikes globally, created by the International Whaling Commission Conservation Committee, is a good example of projects that contribute to and support the development of informed shipping regulations (Schoeman et al., 2020).

In recent years, agent-based models have increasingly been used in fields such as resource management, species interactions, and ecological conservation (McLane et al., 2011). These models consist of computational simulation tools that offer a bottom-up approach to

understanding a system's properties (McLane et al., 2011). By constructing realistic agents and environments, an agent-based model allows for a simulation of all possible interactions between components of a system. These models also allow agents to adapt and learn from experience, thus creating a more accurate representation of reality (McLane et al., 2011). Considering the complex and adaptive nature of human-wildlife interactions, agent-based models can become a prospective tool for analyzing related issues. A study conducted by Anwar et al. (2007) on the impacts of whale-watching tour boats on whales in the St. Lawrence Estuary in Quebec shows the potential of agent-based models for understanding human-wildlife interactions and the negative impacts on marine mammals fostered by anthropogenic activities. This study, alongside numerous others, opens new pathways or informed conservation practices by integrating geospatial data with modeling technologies.

In this light, this research project seeks to build an agent-based model simulating whale ship strike scenarios in a general context. By simulating the behavior of whales, krill, and ships, this model seeks to understand how these agents interact with one another. This primary model will not be an accurate representation of reality. Instead, the main goal of building this model is to understand the process of building an agent-based model. This primary model could eventually be adapted to represent specific system characteristics and spatial contexts, such as vessel frequency, geographic location, and species abundance, thus allowing for a true assessment of whale-ship collision risks for different shipping routes. This paper will present the methods used to build the agent-based model, as well as the results obtained from the simulated model. The model will allow us to see how varying levels of ships affect the whale population.

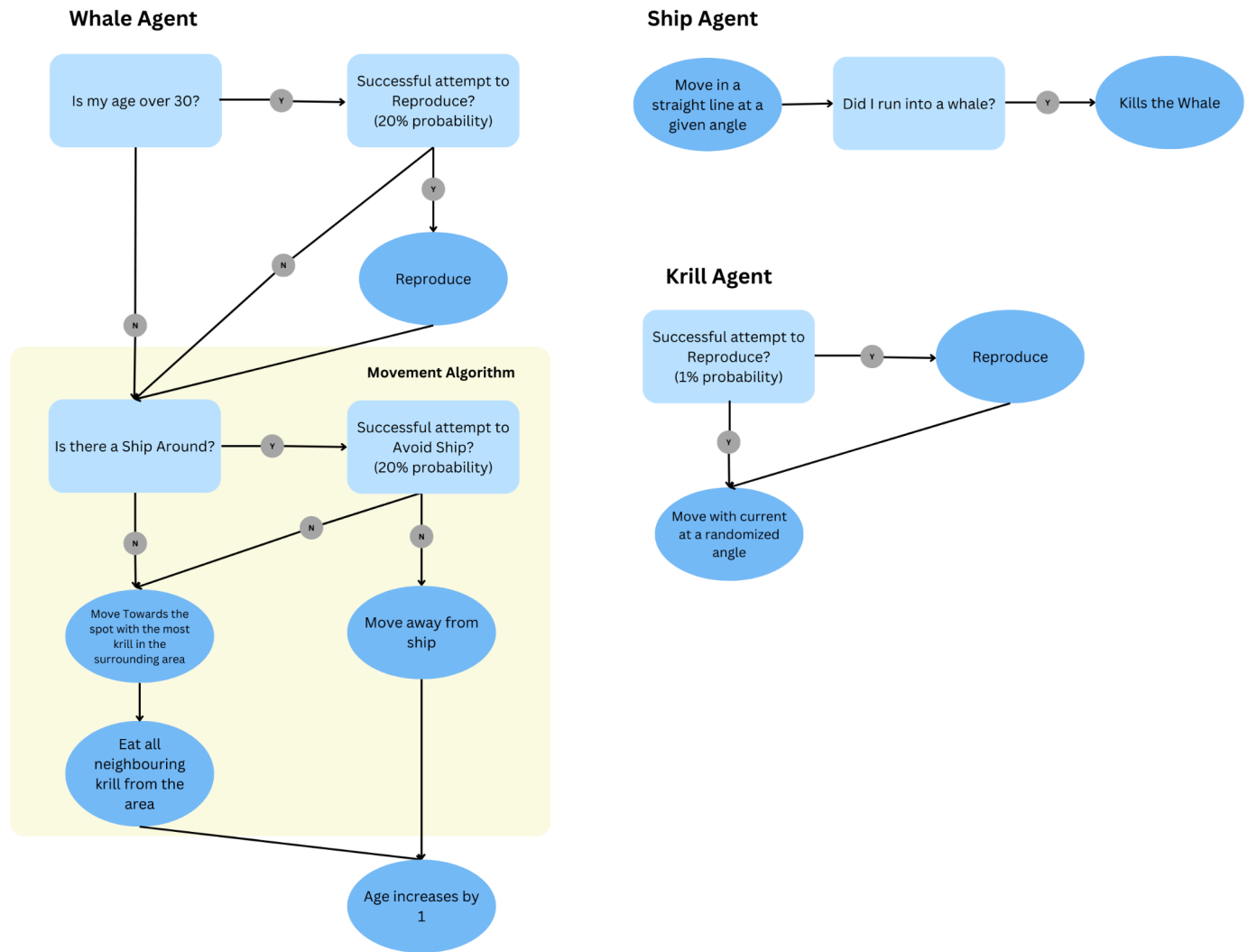
## **Methods and Model Rules**

To build the model, we used Repast Simphony, an agent-based model platform. The model's environment was represented by a 2-dimensional 50x50 unit grid generated by Repast. The grid is not associated with any specific geospatial data, like coordinates. The grid is simply a general representation of space where the agents move and interact with each other. Whales, ships, and krill were the agents in our model. Each agent type was programmed with specific behaviors and attributes:

1. Whale Agent: Programmed to avoid ships, seek high krill density areas, and periodically reproduce (when they have reached a mature age).
2. Krill Agent: Programmed to move with randomized currents and reproduce at a constant rate. Their density is influenced by whale feeding.
3. Ship Agent: Programmed to move in a linear direction at a randomly generated angle.

While simplified, the interactions between these agents were meant to mirror potential real-world behaviors in a synthetic environment. Figure 1 below shows how agents behave and interact with the environment and with each other.

Over discrete time jumps (ticks), the model simulates the movement and interaction of agents. During simulation runs, data such as whale population over time is collected. This data is visualized to understand the patterns and outcomes of the simulations. The simulation will be run multiple times with various numbers of ships to see how ship density affects whale populations.

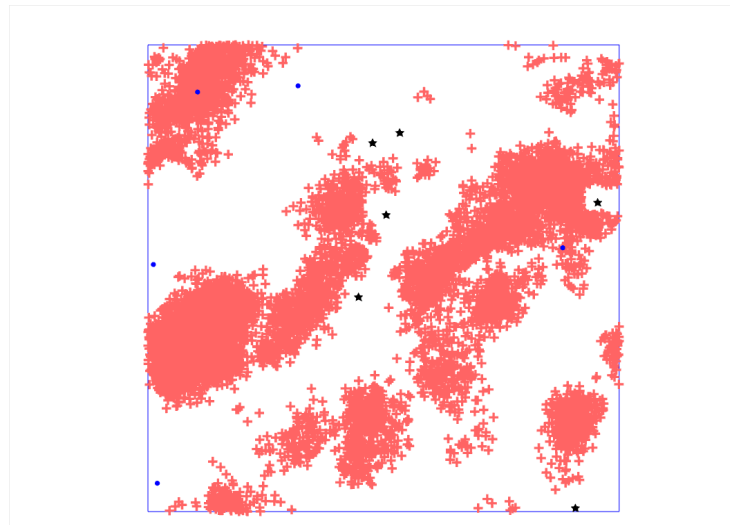


**Figure 1:** ABM model rules for whale, ship, and krill agents.

## Results and Discussion

Results from the agent-based model revealed insights into the interactions among krill, whales, and ships in the simulated environment. A total of three different simulations were conducted using different model parameters for the agents. However, these simulations all shared controlled characteristics. The model was initiated with a constant krill and whale count of 1,000 and 5 respectively, while ship count was modified across the three simulations. All simulations ran for 500 ticks (timestamps). The krill agents were represented by red crosses, the whale agents

by black stars, and the ship agents by blue circles (Figure 2). In all simulations, krill moved according to randomized ocean currents. The whale agents moved toward the neighboring area with the most krill density, consuming all neighboring krill along their path. The dynamic interaction between whales and krill is evident through the fluctuating krill populations. As the whale population increased, the krill population decreased, and vice-versa. The krill population was replenished through periodic reproduction. Ships moved in randomly generated linear patterns. When a ship intersects a whale agent, the whale disappears from the simulation, indicating a fatal injury from a whale-ship collision. This interaction between whales and ships allows for a quantification of the impact of ship traffic on whale population. In total, three simulations with varying ship counts were performed to observe how the whale populations would vary.

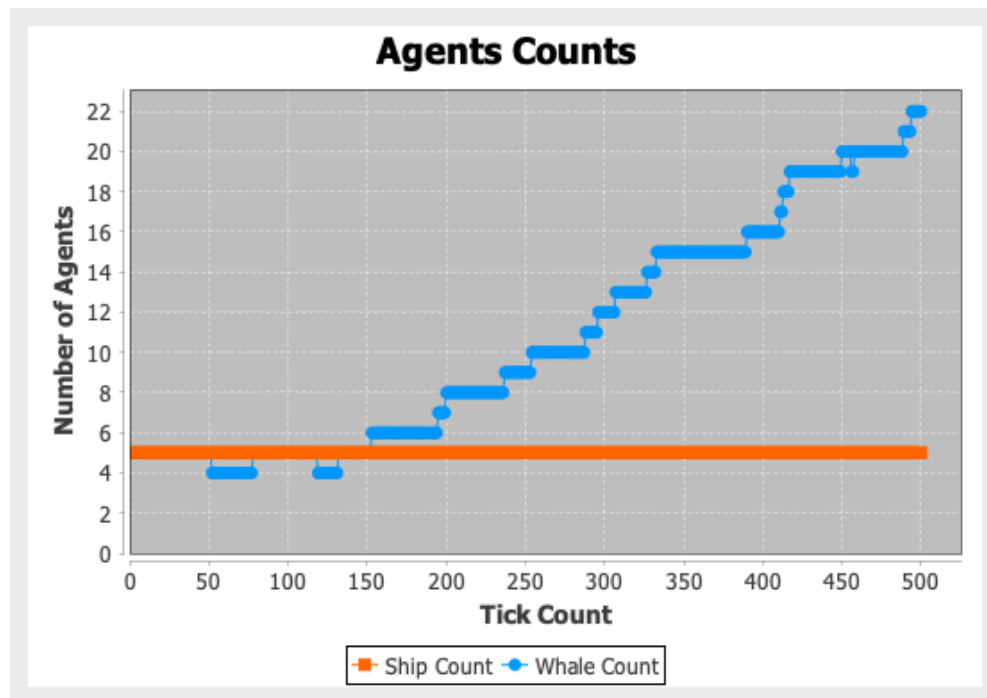


**Figure 3:** The graphical user interface for our ABM after a 500-tick run. The red crosses represent krill, the black stars are whales, and the blue dots are ships.

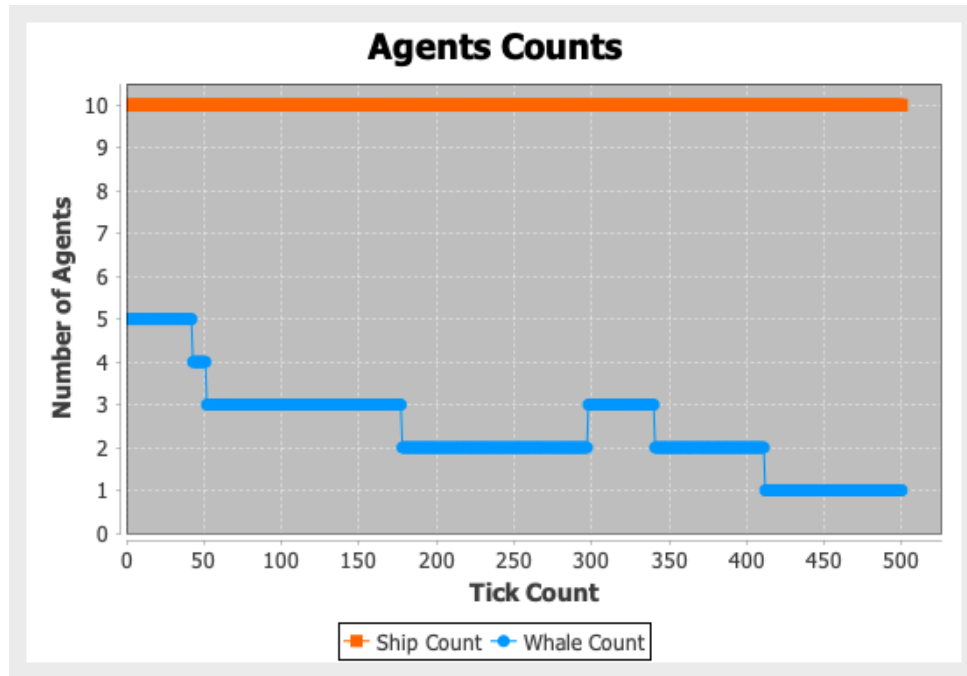
In the first simulation (see Figure 3), the model parameters were set at 5 ships, 10 whales, and 1,000 krill. This simulation exhibited a smaller impact on the whale population in

comparison to the two other simulations. The whale population steadily increased from 5 to 22 whales in a timestamp of 500. In the second simulation (see Figure 4), the model parameters were set at 10 ships, 10 whales, and 1,000 krill. The whale population steadily decreased, reaching a total of 1 by the end of the simulation. Interestingly, at tick 300, a population fluctuation can be observed. A little over halfway through the simulation, the whale population increases by 1 just before decreasing again until the end of the simulation. In the third simulation (see Figure 5), the model parameters were set at 15 ships, 10 whales, and 1,000 krill. This simulation sees a steep decline in whale population, reaching 0 after only 150 ticks.

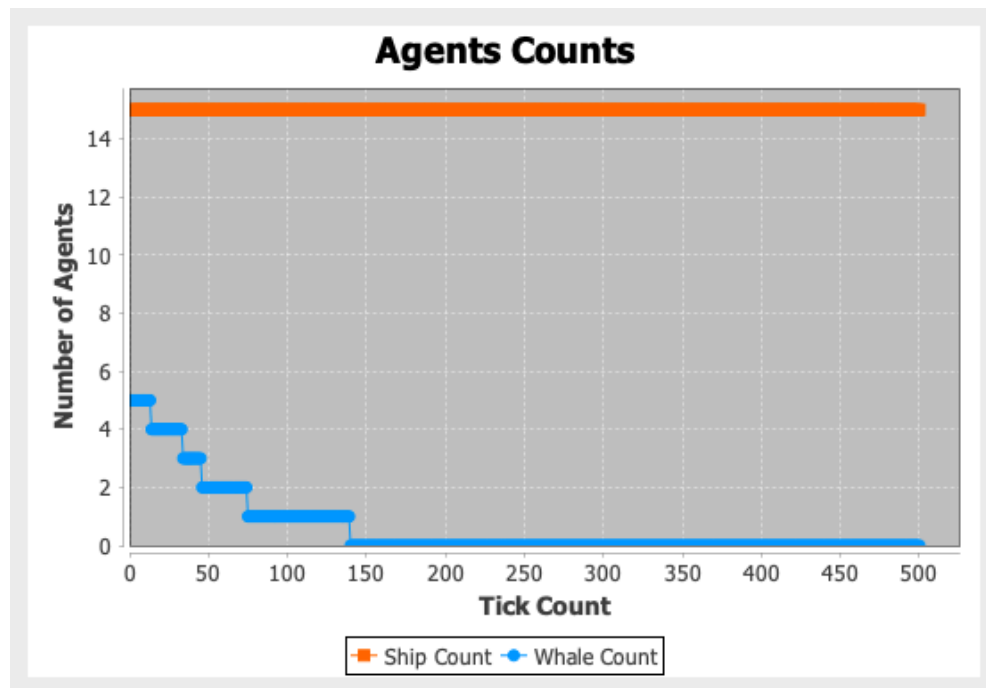
These results indicate that lower ship frequency supports a thriving whale population, while a higher ship frequency can significantly decrease the population. Furthermore, the results from the second simulation suggest that moderating ship frequency may reduce impacts on whale populations. Perhaps a ship count between 5 and 10 may have resulted in a stable whale population across time.



**Figure 3:** Graph showing whale population count with a ship count of 5, over 500 ticks.



**Figure 4:** Graph showing whale population count with a ship count of 10, over 500 ticks.



**Figure 5:** Graph showing whale population count with a ship count of 15, over 500 ticks.

## **Limitations**

While the agent-based model provides a primary framework for simulating whale-ship collision dynamics, it is not without limitations. First, the absence of averaged results from multiple simulations reduces the reliability of the findings. The limited number of simulations conducted inhibits our ability to accurately determine the variance in the results, potentially masking the true variability or consistency of the system's behavior. This limitation is noticeable when performing and comparing various simulations with equal model parameters—the outcomes are always slightly different.

Another limitation is the model's reliance on synthetic data rather than real data. The results derived from the model are limited in their potential to accurately depict reality since real data was not employed. To enhance the model's accuracy, geospatial and ecological data, such as GPS locations of vessels and true krill reproduction rates, can be utilized. Lastly, the complexity of the model can continue to be improved to better represent reality. This could involve adding more attributes to agents. For instance, introducing energy levels for whales or implementing complex ocean currents.

## **Conclusion**

The agent-based model developed in this project allows us to better understand the dynamics of whale-ship collisions. Our simulations, although based on synthetic data, provide an insight into how varying ship frequencies can impact whale populations. The model demonstrates that lower ship frequencies tend to support healthier whale populations, and higher frequencies lead to significant declines. These findings could be important for marine regulatory bodies seeking to balance the needs of maritime commerce with the preservation of marine life.



This project underscores the potential of agent-based models in environmental research and management. By simulating complicated systems, models like these allow us to explore scenarios that are hard to manipulate in the real world. Future projects could vary more parameters such as routes, currents, and krill population, to mimic specific contexts. The identification of high-risk areas and the evaluation of alternative shipping routes, speed limits, and ship density regulations in future studies could lead to more effective mitigation strategies, reducing the risk of fatal ship strikes on whales.

### Works Cited

- Anwar, Sk. M., Jeanneret, C. A., Parrott, L., & Marceau, D. J. (2007). Conceptualization and Implementation of a Multi-agent Model to Simulate Whale-watching Tours in the St. Lawrence Estuary in Quebec, Canada. *Environmental Modelling & Software*, 22(12), 1775-1787. <https://doi.org/10.1016/j.envsoft.2007.02.007>
- International Maritime Organization. (n.d.). *Introduction to IMO*.  
<https://www.imo.org/en/About/Pages/Default.aspx>
- International Maritime Organization. (n.d.). *Marine Environnement*.  
<https://www.imo.org/en/OurWork/Environment/Pages/Default.aspx>
- McLane, A. J., Semeniuk, C., McDermid, G. J., Marceau, D. J. (2011). The Role of Agent-based Models in Wildlife Ecology and Management. *Ecological Modelling*, 222(8), 1544-1556.  
<https://doi.org/10.1016/j.ecolmodel.2011.01.020>
- Schoeman, R. P., Patterson-Abrolat, C., & Plön, S. (2020). A Global Review of Vessel Collisions With Marine Animals. *Frontiers in Marine Science*, 7(292).  
<https://doi.org/10.3389/fmars.2020.00292>